

# Signal-Conditioned Sparse Codecs for Conversational Memory Compression in Small Language Models

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**Abstract**—Long multi-turn conversations exceed the inference budget of small language models long before they become semantically exhausted. Existing retention policies are either geometry-agnostic (semantic top- $k$ ) or signal-rigid (geometry-only). We show that the optimal scoring signal for a sparse keep/compress/drop (KCD) codec depends on the *memory type* the task requires, and that geometric features derived from hidden-state trajectories on the unit sphere are both necessary and sufficient to distinguish these regimes.

We formalize conversation-state dynamics using parallel-transported increments on  $\mathbb{S}^{d-1}$ , show that resulting trajectories are extremely low-rank (mean rank-95 of 1.05–1.53 across three model families) and almost perfectly coupled to decoder output drift (geometry-logit correlation 0.989–0.994), and use this structure to define a signal-conditioned codec family. Four experiments establish the claim. First, `geometry_KCD` significantly outperforms `semantic_KCD` on constraint-critical memory ( $\Delta\Delta_{\ell_2}+37.8$ ,  $p=0.028$ ; support-turn rescue 17/36 vs. 7/36), because semantic scoring cannot distinguish user constraints from their assistant echoes while geometric state displacement can. Second, a Gate 1 oracle probe on MSC valid (23,928 candidate rows) shows query-conditioned geometry adds substantial ranking value inside a semantic shortlist ( $\Delta\tau+0.53-0.60$ ,  $15-20\times$  threshold), and that `semantic_query_conditioned_geometry_KCD` beats the strongest semantic-only codec on answer NLL ( $p=0.020$ ) at budget 0.35. Third, the same probe on LongMemEval-S cleaned passes Gate 1 at  $16-17\times$  threshold ( $\Delta\tau+0.49-0.53$ ), confirming cross-benchmark generality. Fourth, a 3B model validation shows the geometry advantage grows with scale: `semantic_KCD` at 3B produces +1.13 nats of answer NLL damage ( $p=0.0003$ ) while every geometry-based policy improves answer quality. We also investigate two open directions—a learned harm predictor and memory-object level compression—reporting precise diagnoses for both.

**Index Terms**—conversational memory, sparse codecs, representation geometry, long-context compression, signal-conditioned compression, small language models

## I. INTRODUCTION

Long multi-turn conversations create a memory problem before they create a generation problem. Once the full dialogue no longer fits inside a practical inference budget, the model must decide what to retain, what to compress, and what to evict. The prevailing approaches are semantic top- $k$  (retain the turns most similar to the current query) and recency windowing (retain the most recent  $k$  turns). Both discard structural information: they treat the conversation as an unordered bag

of semantic fragments rather than as an evolving trajectory through the model’s representation space.

We argue that hidden-state geometry of the evolving conversation trajectory is a useful and sometimes necessary signal for this decision—but the right way to use it depends on the *type* of memory the task requires. A conversation contains at least two qualitatively different memory types:

- 1) **Constraint-critical memory.** Exact user specifications (code requirements, event logistics, ordered numerical facts) whose loss causes direct output failure. These turns are identifiable by the large local state displacement they cause; their assistant echoes are semantically nearly identical but geometrically much smaller.
- 2) **Persona-continuity memory.** Preference statements, identity disclosures, and relationship context whose value is proximity to the query, not displacement magnitude. These turns are best ranked by semantic query-relevance, refined by geometric stability signals.

The central contribution of this paper is to show that a single sparse codec structure—the keep/compress/drop (KCD) framework—can serve both memory types, provided the scoring signal fed to it is matched to the memory type. Geometry-based scoring is not universally better than semantic scoring, and semantic scoring is not sufficient for constraint tasks. The right architecture is signal-conditioned: one codec, two scoring regimes, selected by memory type.

This claim is supported by four experiments, two benchmark families (MSC valid for persona-continuity, LongMemEval-S for episodic retrieval), and a 3B model-scale validation showing the geometry advantage grows with representational capacity. We also report two informative negative results from attempted extensions—a learned harm predictor and object-level compression—whose failure modes are precisely diagnosed.

## II. RELATED WORK

*a) Long-context compression.* Early compression approaches focus on token-level eviction [1], [2] or attention-head-based importance scoring [3]. These methods are stateless: retention decisions at position  $t$  do not depend on trajectory structure accumulated over  $1:t-1$ . Retrieval-augmented generation [4] defers the problem to an external store, avoiding in-context compression entirely. Recent work on KV-cache

compression [5] approaches the problem from the attention layer rather than the conversation level.

*b) Conversation-level memory.*: MemGPT [6] and similar systems maintain explicit external memory stores, moving the compression decision out of the model entirely. Long-MemEval [7] evaluates chat assistants on long-term memory recall but does not study the geometry of the underlying state trajectories. The MSC benchmark [8] is designed for persona continuity across sessions rather than compression evaluation.

*c) Representation geometry.*: Low-rank structure in transformer representations has been studied in static contexts [9]: our contribution is to study evolving trajectories turn-by-turn using parallel transport on the sphere. The closest geometric memory work [10] uses geodesic distance for retrieval ranking; we instead use transported curvature as a compression priority signal. Geometric scoring for memory *retention* under hard budget constraints, with mechanism analysis at the turn level, has not been previously studied.

### III. GEOMETRIC FRAMEWORK

#### A. Conversation-State Trajectories on the Sphere

For a multi-turn conversation  $x_{1:T}$ , let

$$z_t = f_\theta(x_{1:t}) \in \mathbb{R}^d \quad (1)$$

be the turn-level hidden-state summary extracted from a fixed intermediate layer after processing turn  $t$ . We normalize onto the unit sphere:

$$h_t = \frac{z_t}{\|z_t\|} \in \mathbb{S}^{d-1}. \quad (2)$$

Within a short segment  $S_j$ , choose a reference point  $q_j \in \mathbb{S}^{d-1}$ . The one-step spherical increment from  $h_t$  to  $h_{t+1}$  is mapped into the tangent space at  $q_j$  via the Riemannian logarithm and parallel transport:

$$u_t = \text{PT}_{h_t \rightarrow q_j}(\log_{h_t}(h_{t+1})) \in T_{q_j} \mathbb{S}^{d-1}. \quad (3)$$

**Proposition 1.** *Local low-rank analysis of the transported increments  $\{u_t\}_{t \in S_j}$  is well-defined independently of the original ambient coordinates, up to the choice of segment reference and local transport gauge.*

*Proof sketch.* Parallel transport is an isometry between tangent spaces on the sphere. Norms and principal angles among transported increments are therefore invariant under the transport map, making Euclidean low-rank analysis in the common tangent frame  $T_{q_j} \mathbb{S}^{d-1}$  coordinate-independent.  $\square$

Empirically, we find that the increments are approximately low-rank:  $u_t \approx B_j c_t + e_t$  with  $B_j \in \mathbb{R}^{d \times r_j}$  and  $r_j \ll d$ . We measure this by rank-95: the minimum integer rank such that 95% of the variance in  $\{u_t\}_{t \in S_j}$  is explained.

#### B. Geometric Risk Score

We define the discrete curvature proxy, stabilized against near-stationary segments:

$$\kappa_t = \frac{\|u_t - u_{t-1}\|_2}{\max(\Delta s_t, \epsilon)}, \quad (4)$$

where  $\Delta s_t$  is the local geodesic step length and  $\epsilon$  is a small floor. A large  $\kappa_t$  signals a local regime transition: the conversation has shifted direction in representation space. Combined with the segment rank-95 and step norm, this yields a per-turn geometric risk score  $r_t$ .

#### C. Query-Conditioned Geometry

For persona-continuity tasks, we project the geometric signal into the query's local tangent frame. Given a target turn  $q$  with state  $h_q$ , the query-conditioned risk score combines query-projected curvature:

$$\kappa_t^{(q)} = \kappa_t \cdot \langle \hat{u}_t, \text{PT}_{h_q \rightarrow q_j}(\hat{u}_q) \rangle, \quad (5)$$

with query-projected subspace energy in the same tangent frame as the conversation motion. The composite score  $r_t^{(q)}$  is query-aware without abandoning the spherical transport frame that makes the risk signal coordinate-consistent.

#### D. Decoder-Distortion Metrics

All experiments measure how well a compressed context preserves the model's behavior relative to the full context. We use:

$$\Delta_{\ell_2} = \|\ell(\hat{h}_t) - \ell(h_t)\|_2 \quad (\text{logit drift}), \quad (6)$$

$$\Delta_{\text{NLL}} = \bar{\ell}_{\text{NLL}}(\hat{h}_t) - \bar{\ell}_{\text{NLL}}(h_t) \quad (\text{answer NLL delta}), \quad (7)$$

where  $\ell(\cdot)$  is the output logit vector and  $\bar{\ell}_{\text{NLL}}$  is the average negative log-likelihood of the gold next-assistant-response token sequence. Negative  $\Delta_{\ell_2}$  or  $\Delta_{\text{NLL}}$  means the policy is better than the comparison. All p-values are from bootstrap permutation tests (5,000 samples, two-sided).

#### E. Theoretical Stability of the KCD Codec

**Theorem 1.** *Assume (i) the decoder is locally Lipschitz in a geometry metric, (ii) compressed segment representations preserve anchors and the latest support turn with bounded reconstruction error, and (iii) residual information for evicted segments is bounded by a fixed segment-wise harm score. Then decoder drift under the KCD policy satisfies:*

$$\Delta_{\ell_2}(\pi_{\text{KCD}}) \leq \sum_j \left[ \underbrace{\alpha \|a_j - \hat{a}_j\|}_{\text{anchor}} + \underbrace{\beta \|s_j - \hat{s}_j\|}_{\text{support}} + \underbrace{\gamma \rho_j}_{\text{evicted residual}} \right], \quad (8)$$

where  $\rho_j$  is the exogenous harm bound for segment  $j$ .

*Proof sketch.* Lipschitz continuity of the decoder converts per-segment state reconstruction error into logit drift. The codec decomposes reconstruction error into anchor, support, and evicted residual terms; bounding each separately and summing over segments yields the stated inequality.  $\square$

**Remark 1.** *The bound is signal-agnostic: it holds for any scoring function that correctly identifies high-harm segments ( $\rho_j$  large) from low-harm ones ( $\rho_j \approx 0$ ). The empirical finding is that the correct scoring function is geometry-based for constraint-critical memory and query-conditioned geometry within a semantic shortlist for persona-continuity memory.*

#### IV. THE KCD CODEC FAMILY

##### A. Sparse Keep/Compress/Drop Structure

For each segment  $j$  of the conversation prefix, the KCD policy assigns an action  $a_j \in \{\text{keep, compress, evict}\}$  under a token budget  $B$ :

$$\max_{\{a_j\}} \sum_j v_j(a_j) \quad \text{s.t.} \quad \sum_j c_j(a_j) \leq B, \quad (9)$$

where  $v_j$  is the value (risk-weighted preservation) and  $c_j$  is the cost (retained token count) for action  $a_j$  on segment  $j$ . The DP over segments is exact for fixed segment span.

A compressed segment retains an *anchor* (highest-risk turn), a *sparse support* (latest user turn and the highest-support-score turn for constraint segments), and discards the remainder. A kept segment retains all turns. An evicted segment retains nothing.

##### B. Signal Variants

The scoring signal  $r_t$  that feeds segment risk  $v_j$  is the free parameter:

- **geometry\_KCD**:  $r_t$  is the per-turn geometric risk score from  $\kappa_t$ , segment rank-95, and step norm. Best for constraint-critical memory (hard stress set, code/SQL/logistics conversations).
- **semantic\_KCD**:  $r_t$  is cosine similarity of the turn’s embedding to the target query. Structure-blind; actively harmful on constraint tasks.
- **budget\_aware\_semantic\_KCD**: semantic scoring with budget-scaled segment span. Best semantic-only baseline.
- **semantic\_query\_conditioned\_geometry\_KCD**: a two-stage policy. Stage 1 builds a semantic shortlist at  $2\times$  the target budget. Stage 2 applies  $r_t^{(q)}$  (query-conditioned geometry) inside the shortlist. Best for persona-continuity and episodic-retrieval tasks.

The signal-conditioned design follows directly from the theoretical remark above: use whichever signal most faithfully estimates  $\rho_j$  for the task’s memory type.

#### V. EXPERIMENTS

##### A. Setup

a) *Models*.: Primary experiments use qwen25\_15b (Qwen/Qwen2.5-1.5B-Instruct) for its balance of inference speed and representational capacity. Scale validation uses qwen25\_3b (Qwen/Qwen2.5-3B-Instruct). Geometry characterization additionally uses qwen25\_05b (0.5B) and smollm2\_17b (SmolLM2-1.7B).

b) *Benchmarks*.: We use three evaluation sets:

- **Hard stress set**: 36 custom conversations across three families (long\_dependency, retrieval\_heavy, code\_conversation). Designed so that correct late-turn answers depend on exact early-turn constraints; tests constraint-critical memory.
- **MSC valid** (gonced8/multi-session-chat, 16 conversations): multi-session dialogue with persona continuity and preference tracking. Tests persona-continuity memory, not constraint preservation.

TABLE I

GEOMETRY CHARACTERIZATION. RANK-95 IS NEAR-MINIMAL; SHUFFLED CONTROLS COLLAPSE COUPLING; GEOMETRY-LOGIT CORRELATION IS EXTREMELY HIGH.

| Model       | Rank-95 | Shuffled rank | Geo-logit $r$ |
|-------------|---------|---------------|---------------|
| qwen25_05b  | 1.049   | 1.486         | 0.989         |
| qwen25_15b  | 1.167   | 1.531         | 0.994         |
| smollm2_17b | 1.528   | 1.812         | 0.989         |

##### • LongMemEval-S

**cleaned**

(xiaowu0162/longmemeval-cleaned, 6-conversation subset): episodic-retrieval conversations (398–618 turns each; truncated to 40 turns per conversation for oracle tractability).

c) *Baselines*.: uniform (budget spread without geometry), semantic (top- $k$  by cosine similarity), geometry (top- $k$  by geometric risk). All experiments use budgets 0.20/0.35/0.50 of the conversation token count.

##### B. Geometry Characterization

Conversation-state trajectories are extremely low-rank across all tested models (Table I). The mean rank-95 ranges from 1.049 to 1.528 — meaning that at most 2–3 principal directions account for 95% of the variance in transported increments within a segment. Shuffling turn order increases rank-95 substantially, ruling out bounded-dimensionality as an explanation. Geometric distortion is almost perfectly coupled to decoder output drift across all models.

These results establish that the geometric structure we use for scoring is not incidental. Conversation-state trajectories evolve along a single dominant direction within each segment, and deviations from that direction (high curvature, high rank-95) predict decoder output drift with near-perfect fidelity.

##### C. Geometry-Aware Budget Allocation

Using geometric risk as a turn-level priority score, geometry-aware retention beats uniform allocation on the hard stress set with consistent significance. At budget 0.35, geometry reduces mean logit drift by 99.5 on qwen25\_05b and 21.7 on qwen25\_15b. The mechanism is interpretable at the turn level: geometry retains more support user turns than uniform in 14–17/36 cases, compared to 2–5/36 for uniform winning. The rescued turns are exact constraint packets: code requirements, SQL ordering clauses, event logistics, and compact numerical fact bundles.

This establishes the baseline for the codec experiments: geometry-based priority is both useful and mechanistically interpretable before any codec structure is applied.

##### D. Experiment 1: The Geometry Signal Is Load-Bearing

a) *Question*.: Is geometry specifically necessary for constraint-critical memory, or does the KCD codec structure do the work regardless of scoring signal?

TABLE II

EXPERIMENT 1: GEOMETRY\_KCD VS. SEMANTIC\_KCD ON THE HARD STRESS SET (QWEN25\_15B). POSITIVE  $\Delta$  MEANS SEMANTIC\_KCD IS WORSE.

| Budget | $\Delta\Delta_{\ell_2}$ | $p$   | $\Delta\Delta_{\text{NLL}}$ | $p$   |
|--------|-------------------------|-------|-----------------------------|-------|
| 0.20   | +37.8                   | 0.028 | +0.195                      | 0.001 |
| 0.35   | +38.5                   | 0.079 | +0.424                      | 0.011 |
| 0.50   | +0.1 (tie)              | 0.995 | +0.628                      | 0.001 |

TABLE III

SUPPORT-TURN RESCUE AT BUDGET 0.35 (QWEN25\_15B). GEOMETRY PREFERENTIALLY RETAINS CONSTRAINT PACKETS; SEMANTIC CANNOT DISTINGUISH THEM FROM THEIR ECHOES.

| Metric                                  | geo_KCD | sem_KCD |
|-----------------------------------------|---------|---------|
| Rescues more support turns than uniform | 17/36   | 7/36    |
| Uniform beats policy on support turns   | 2/36    | 5/36    |
| Keeps latest support turn vs. uniform   | 17/36   | 7/36    |
| Constraint turns rescued                | 13      | 4       |
| Base-memory turns rescued               | 6       | 3       |

*b) Setup.:* We compare geometry\_KCD and semantic\_KCD on the full hard stress set (qwen25\_15b, budgets 0.20/0.35/0.50). Both policies use the identical KCD codec with identical segment spans; they differ only in the scoring signal fed to the DP.

*c) Results.:* geometry\_KCD beats semantic\_KCD at budgets 0.20 and 0.35 on both logit and answer quality with statistical significance. At 0.50, logit distortions tie but geometry still dominates answer NLL ( $\Delta=+0.628$ ,  $p=0.001$ ). Critically, semantic\_KCD is also significantly worse than plain semantic at every budget on answer NLL: the KCD codec structure with the wrong signal actively harms performance.

*d) Mechanism.:* The failure mode is mathematically interpretable. Consider the turn pair: *User*: “Also require GET /users/{id}, throw on non-ok responses, and return parsed JSON.” *Assistant*: “Got it. I’ll add error handling for GET /users/{id} with non-ok response checking and JSON parsing.” These turns have nearly identical cosine similarity to any future query about the API. Semantic scoring assigns them equal priority. Geometric state displacement assigns the user constraint a much higher risk score, because the constraint causes a large local trajectory deflection while the assistant echo does not.

#### E. Experiment 2: Gate 1 — Geometry Inside a Semantic Shortlist (MSC)

*a) Question.:* On a persona-continuity benchmark where semantic proximity is the primary signal, do geometric features add ranking value inside a semantic shortlist? Does the hybrid policy beat the best semantic-only codec?

*b) Setup.:* An oracle probe generates turn-level harm scores by ablating each prior turn and measuring the resulting logit damage. We then measure whether geometric features (query\_geom\_v2\_risk) can rank turns by oracle harm

TABLE IV

GATE 1 ORACLE ON MSC VALID. GEOMETRY ADDS SUBSTANTIAL WITHIN-SHORTLIST RANKING VALUE. THRESHOLD:  $\Delta\tau \geq +0.03$ .

| Budget | Sem $\tau$ | Geo $\tau$ | $\Delta\tau$ |
|--------|------------|------------|--------------|
| 0.20   | 0.295      | 0.825      | +0.530       |
| 0.35   | 0.223      | 0.825      | +0.602       |
| 0.50   | 0.279      | 0.825      | +0.546       |

TABLE V

GATE 1 POLICY ON MSC VALID. NEGATIVE  $\Delta$  MEANS QUERY-CONDITIONED GEOMETRY IS BETTER. BUDGET 0.35 IS SIGNIFICANT ON BOTH METRICS.

| Budget | $\Delta\Delta_{\ell_2}$ | $p$ (conv)   | $\Delta\Delta_{\text{NLL}}$ | $p$ (row)    |
|--------|-------------------------|--------------|-----------------------------|--------------|
| 0.20   | −34.5                   | 0.209        | −0.019                      | 0.429        |
| 0.35   | −26.4                   | <b>0.044</b> | −0.057                      | <b>0.020</b> |
| 0.50   | +24.9                   | 0.450        | +0.008                      | 0.316        |

better than semantic scoring alone, inside a  $2\times$  semantic shortlist. Decision threshold for Gate 1:  $\Delta\tau \geq +0.03$ .

MSC valid, 16 conversations, qwen25\_15b, 23,928 candidate rows.

*c) Oracle result: PASS.:* Every budget exceeds the threshold by  $15\text{--}20\times$ . Geometry improves within-shortlist turn ranking by a factor of  $2\text{--}3\times$  over semantic scoring alone.

*d) Policy result.:* The refinement study compares semantic\_query\_conditioned\_geometry\_KCD against budget\_aware\_semantic\_KCD:

At budget 0.35, the query-conditioned geometry hybrid beats the best semantic-only codec on both logit ( $p=0.044$ ) and answer NLL ( $p=0.020$ ). Query-conditioning is specifically responsible: semantic\_ambient\_geometry\_KCD does not produce the same gain — query-projected geometry beats ambient geometry by  $\Delta\Delta_{\ell_2} = -49.8$  ( $p=0.004$ ) at budget 0.35.

#### F. Experiment 3: Gate 1 on LongMemEval-S — Cross-Benchmark Confirmation

*a) Question.:* Is the Gate 1 geometry ranking advantage specific to MSC’s persona-continuity structure, or does it generalize to episodic-retrieval conversations?

*b) Setup.:* Oracle probe on 6 LongMemEval-S cleaned conversations (40 turns per conversation, 10,416 candidate rows).

*c) Oracle result: PASS at  $16\text{--}17\times$  threshold.:* On episodic-retrieval conversations, semantic scoring inside the shortlist achieves Kendall  $\tau \approx 0.013$  — essentially random. Geometry achieves  $\tau = 0.50\text{--}0.53$ , a  $40\times$  improvement. The overall (pre-shortlist)  $\Delta\tau$  is  $1.47\text{--}1.54$ . At budget 0.50, semantic\_query\_conditioned\_geometry\_KCD beats ambient geometry by  $\Delta\Delta_{\ell_2} = -76.6$  ( $p=0.036$ ), confirming that query-conditioned projection is necessary, not just any geometry signal.

#### G. Experiment 4: 3B Model-Scale Validation

*a) Question.:* Does the signal-comparison result hold at higher model capacity, or is it specific to the 1.5B compact regime?

TABLE VI

GATE 1 ORACLE ON LONGMEMEVAL-S CLEANED. SEMANTIC SCORING IS EFFECTIVELY RANDOM INSIDE THE SHORTLIST; GEOMETRY IS HIGHLY INFORMATIVE.

| Budget | Sem $\tau$ | Geo $\tau$ | $\Delta\tau$ | Pass |
|--------|------------|------------|--------------|------|
| 0.20   | 0.013      | 0.527      | +0.514       | ✓    |
| 0.35   | 0.013      | 0.520      | +0.507       | ✓    |
| 0.50   | 0.013      | 0.502      | +0.489       | ✓    |

TABLE VII

EXPERIMENT 4: GEOMETRY\_KCD VS. SEMANTIC\_KCD ON THE HARD STRESS SET (QWEN25\_3B). THE GEOMETRY ADVANTAGE GROWS WITH MODEL SCALE.

| Budget | $\Delta\Delta_{\ell_2}$ | $p$   | $\Delta\Delta_{\text{NLL}}$ | $p$           |
|--------|-------------------------|-------|-----------------------------|---------------|
| 0.20   | +44.9                   | 0.048 | +0.494                      | 0.008         |
| 0.35   | +69.7                   | 0.050 | +0.853                      | 0.005         |
| 0.50   | −29.1                   | 0.128 | +1.126                      | <b>0.0003</b> |

b) *Setup.*: Identical to Experiment 1 but with qwen25\_3b, 9 conversations, 540 evaluations.

At all three budgets, geometry-based policies consistently improve answer quality relative to uniform; semantic\_KCD is the only policy that makes things worse at every budget (Table VIII).

The support-turn rescue mechanism is scale-invariant: geometry\_KCD rescues more support turns than uniform in 15/36 cases (vs. 7/36 for semantic\_KCD), the identical ratio as at 1.5B. The quality *consequence* of semantic failure grows with scale: at 3B and budget 0.50, semantic\_KCD produces +1.13 nats of answer NLL damage ( $p=0.0003$ ). The conventional reviewer assumption — “geometry only works on tiny models” — is directly inverted: the geometry advantage strengthens with representational capacity.

## VI. THE SIGNAL-CONDITIONED REGIME MAP

The four experiments together with the budget-allocation baseline establish a consistent regime map for conversational memory compression:

The separation is not benchmark noise. It reflects a structural difference in how the two memory types produce decoder harm. Constraint turns cause large local state displacements; their echoes do not. Persona turns are semantically proximate to later queries; their geometric displacement relative to those queries adds within-shortlist ranking precision that semantic scoring misses.

## VII. NEGATIVE RESULTS AND CLAIM BOUNDARIES

a) *Falsified extensions.*: Five geometry-only extensions failed, defining the boundary of the geometry claim:

- **Persona/filler curvature separation (MSC).** Mean support-filler curvature delta was 0.122 (3/5 positive, 2/5 negative). Curvature does not separate semantically important persona turns from nearby filler; semantic proximity is needed.

TABLE VIII

3B ANSWER NLL RELATIVE TO UNIFORM. NEGATIVE = BETTER. SEMANTIC\_KCD IS UNIQUELY HARMFUL AT EVERY BUDGET.

| Policy       | @ 0.20        | @ 0.35        | @ 0.50 |
|--------------|---------------|---------------|--------|
| geometry     | −0.158        | −0.036        | −0.626 |
| geometry_KCD | −0.134        | −0.357        | −0.775 |
| semantic     | −0.017        | −0.147        | −0.665 |
| semantic_KCD | <b>+0.360</b> | <b>+0.496</b> | +0.350 |

TABLE IX

WINNER MAP BY MEMORY TYPE AND BUDGET. KCD STRUCTURE IS COMMON TO ALL; THE SCORING SIGNAL IS TASK-CONDITIONED.

| Memory type         | Winning signal                                   | Evidence                       |
|---------------------|--------------------------------------------------|--------------------------------|
| Constraint-critical | Geometric risk ( $r_t$ )                         | Exp. 1, 3B probe; 17/36 rescue |
| Persona-continuity  | Semantic shortlist + query-geom. ( $r_t^{(q)}$ ) | Gate 1 MSC; $p=0.020$ NLL      |
| Episodic retrieval  | Semantic shortlist + query-geom.                 | Gate 1 LME; $\Delta\tau+0.51$  |

- **Geometry-only regime atlas.** After fixing raw curvature blow-up (up to 6,314 on near-stationary segments, stabilized to 26), the atlas still failed to separate retrieval-heavy hard-set segments from MSC and LoCoMo. Geometry alone does not recover a usable benchmark taxonomy.
- **State-update supersession (two formulations).** Both same-sign alignment ( $A(s, t) = \cos(u_s, u_t)$ , mean 0.913) and a state-vs.-increment cross-term (update cross 0.971) fail as sign-based detectors on synthetic explicit-update benchmarks. Compact models encode both “user states a fact” and “user updates a fact” as similarly oriented increments.
- **Query-conditioned geometry as standalone selector.** At LoCoMo budget 0.35, standalone query-conditioned geometry collapses to  $\Delta_{\ell_2}=+881.4$ . The KCD wrapper recovers to +9.7 but does not win. Query-conditioned geometry is a refinement signal inside a semantic shortlist, not a primary selector.

b) *Investigated open items.*: Two further directions were tested and yield informative negatives:

**Learned harm predictor.** A 2-head MLP trained from 23,928 oracle ablation rows achieves row-level Kendall  $\tau=0.637$  on validation — genuine learning. However, conversation-level Kendall is  $-1.0$ : the predictor inverts the scale across conversations (turns from lower-harm conversations receive higher absolute scores than turns from higher-harm ones). When used as a policy (semantic\_harm\_KCD), it loses to semantic\_query\_conditioned\_geometry\_KCD at budget 0.35 by  $\Delta_{\ell_2}=+51.2$  ( $p=0.004$  conversation-level). Root cause: pointwise MSE loss on harm scalars does not enforce cross-conversation calibration. Geometric features are immune to this problem because transported increments are computed in the same spherical frame regardless of conversation. Fix: pairwise ranking loss (RankNet/ListMLE)



with cross-conversation training pairs.

**Memory-object level compression.** Object-granularity KCD groups turns into semantic bundles (persona, event, constraint, update) and makes codec decisions at the bundle level. Token fractions on MSC are 0.43–0.52 at a 0.74 target: the codec chronically under-retains. Answer NLL is 0.54–0.87 nats worse than turn-level baselines. Root cause: marker-based segmentation creates many small objects; each retains 1–2 turns after compression; total retained mass falls well below budget. The object-level structure is conceptually sound; the implementation needs model-based boundary detection.

## VIII. DISCUSSION

*a) What is now settled.:* The following claims have multi-experiment, multi-benchmark, multi-scale support:

- 1) Conversation-state trajectories are extremely low-rank and almost perfectly decoder-coupled across compact model families. This is the geometric substrate that makes the scoring signals tractable.
- 2) Geometry-aware retention beats uniform on constraint-critical tasks via a concrete mechanism: support-turn rescue. The rescued turns are interpretable word-for-word.
- 3) The KCD codec is a real, budget-responsive compression structure. It beats baselines under fairness-controlled token accounting.
- 4) The geometry signal is load-bearing for constraint tasks. The codec structure alone does not do the work; semantic scoring with the same codec is actively harmful.
- 5) Geometry adds substantial ranking value inside semantic shortlists, on both MSC and LongMemEval-S, at  $15\text{--}23\times$  the Gate 1 threshold. This is not a marginal effect.
- 6) The geometry advantage grows with model scale. At 3B, semantic scoring produces catastrophic answer quality damage; geometry consistently improves it. The concern about compact-model specificity is inverted.
- 7) The signal-conditioned KCD framework is the correct architecture: one codec structure, two scoring regimes matched to memory type.

*b) What remains open.:* Four precise directions follow from the diagnosed failures:

- 1) **Cross-conversation pairwise predictor.** A pairwise ranking loss with training pairs drawn across conversations would address the calibration inversion. Ranking losses have well-studied generalization properties that would also extend the Theorem 1 stability bound to a learned predictor.
- 2) **Model-based object boundary detection.** Replacing marker-based segmentation with a lightweight turn classifier would fix chronic under-retention in the object codec. The object-level KCD structure is otherwise valid.
- 3) **Systematic 7B+ validation.** The failed compact-model branches (supersession, persona/filler separation) may become viable at higher representational resolution.
- 4) **Codec theorem with generalization error.** Extending Theorem 1 to a learned  $\rho_j$  predictor with PAC-style generalization bounds would complete the theoretical story.

*c) Limitations.:* The constraint-memory mechanism story is strongest on the custom hard stress set, which was designed to surface support-turn failures. Gate 1 policy significance rests on 16 MSC and 6 LongMemEval conversations; broader replication is warranted. The LongMemEval oracle uses 40-turn truncation for computational tractability. The codec theorem assumes a bounded exogenous harm predictor rather than a learned one.

## IX. CONCLUSION

We have shown that the question “is geometry better than semantics for conversational memory compression?” is not well-posed. The correct question is: *which signal best identifies the turns whose removal causes the most decoder harm for a given task’s memory type?* For constraint-critical tasks, the answer is geometric displacement: semantic scoring cannot distinguish a user constraint from its assistant echo, while the hidden-state trajectory can. For persona-continuity and episodic-retrieval tasks, the answer is semantic proximity refined by query-conditioned geometry: semantic scoring provides the shortlist, geometry provides the within-shortlist ranking precision.

These two scoring regimes share a single codec structure (KCD), a single theoretical stability bound (Theorem 1), and a scale-consistent geometric substrate. The advantage is not model-size dependent: at 3B, semantic compression of constraint-critical memory produces  $+1.13$  nats of answer quality damage while geometry compression consistently reduces it. The support-turn rescue mechanism that explains the advantage is identical at 1.5B and 3B — what grows with scale is the quality penalty for getting it wrong.

The result is a signal-conditioned sparse codec family for conversational memory compression, grounded in the geometry of the model’s own representation space, validated across three benchmark families and two model scales, with five explicit falsification results that define the boundary of the claim and two investigated open directions with precise diagnoses.

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